

Brandon Mossop

Ph.D. Candidate, School of Computing, Queen's University

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Profile

Ph.D. Candidate in Computing at Queen's University and Lead Data Scientist at Venford Technologies. My research spans causal inference, Bayesian analysis, simulation, and explainable artificial intelligence (AI), with a particular focus on causal discovery, forecasting, and decision-making under changing conditions. I am increasingly interested in extending these ideas to causal world models and physical AI.

Education

Queen's University

Kingston, Ontario

Ph.D. in Computing, expected 2026

2023–2026

- Ph.D. Thesis: *Causal Structure Learning for Robust and Explainable Artificial Intelligence*.
- Supervisor: Dr. Salimur Choudhury.
- Graduate researcher, GOAL Lab, School of Computing.

Lakehead University

Thunder Bay, Ontario

M.Sc. in Computer Science, Specialization in Artificial Intelligence

2020–2022

- Master's Thesis: *Hybrid Deep Learning with Stacked Dilated Causal Convolutions for Health Forecasting using Multivariate Time-series Data*.
- Supervisor: Quazi Abidur Rahman

Lakehead University

Thunder Bay, Ontario

Honours B.Sc. in Computer Science

2017–2020

Lakehead University

Thunder Bay, Ontario

B.Eng. in Civil Engineering

2010–2014

Peer-Reviewed Publications

Journal Articles

1. **Mossop, B.**, & Choudhury, S. (2026). Regime-aware causal Bayesian forecasting for non-stationary time series. *PLOS ONE*, 21(8), e0356474. DOI
2. **Mossop, B.**, & Choudhury, S. (2026). Improving machine learning forecasting explainability and performance with causal feature selection for infectious disease spread. *INFOR: Information Systems and Operational Research*. DOI

Conference Papers

3. **Mossop, B.**, & Choudhury, S. (2026). Causal Story-Guided Agentic Causal Discovery with Small Language Models. **Accepted** to the 4th International Conference on Foundation and Large Language Models (FLLM 2026).
4. **Mossop, B.**, & Choudhury, S. (2025). Agent-guided Causal Discovery with a Small Language Model. *2025 3rd International Conference on Foundation and Large Language Models (FLLM)*, 693–698. DOI
5. **Mossop, B.**, & Rahman, Q. A. (2023). Infectious Disease Forecasting using Multivariate Incomplete Time-series: A Hybrid Architecture with Stacked Dilated Causal Convolutions. *2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 4220–4227. DOI
6. **Mossop, B.**, Bismil, R., & Rahman, Q. A. (2023). Forecasting Hospital Mental-Health Admissions with a Novel Hybrid Deep Learning Architecture. *2023 IEEE International Conference on Bioinformatics*

Doctoral Research Experience

GOAL Lab, Queen's University

Kingston, Ontario

Graduate Researcher

2023–Present

- Develop causal discovery and causal forecasting methods for time-series systems.
- Supervise undergraduate research assistants, providing guidance on research design, implementation, experimentation, and technical writing for projects in causal AI and machine learning.
- Designed *CaReBayes*, a regime-aware Bayesian forecasting framework combining temporal causal discovery, regime identification, Bayesian structural autoregression, and posterior predictive uncertainty.
- Developed causal feature-selection methods for infectious-disease forecasting, using learned causal structure to improve interpretability and predictive performance.
- Investigate agent-guided causal discovery using small language models, including agent roles for orientation, edge augmentation, critique, and causal story reasoning.
- Conduct reproducible experiments involving causal graphs, interventions, uncertainty quantification, and statistical and machine-learning baselines.

Selected Research Contributions

Causal Forecasting and Non-Stationarity

- Integrated temporal causal structure learning with Bayesian forecasting to model regime-dependent dynamics rather than assuming a fixed data-generating process.
- Used posterior predictive distributions to quantify forecast uncertainty and structural causal models to support intervention-oriented forecasting.

Agentic Causal Discovery

- Studied how small language models can contribute contextual knowledge to causal discovery while statistical conditional-independence tests provide data-driven constraints.
- Developed multi-agent workflows for graph orientation, augmentation, critique, and causal explanation.

Health and Biomedical Forecasting

- Developed deep-learning and causal-ML approaches for infectious-disease and hospital-demand forecasting using multivariate time-series data.

Research and Professional Experience

Venford Technologies

Canada

Lead Data Scientist

2026–Present

- Develop modelling, simulation, and machine-learning methods for decision intelligence and digital-twin applications.
- Contribute to the core technical architecture underlying the modelling and simulation platform.
- Implement computational approaches for scenario analysis, uncertainty quantification, policy evaluation, and complex-system decision-making.
- Translate research ideas in probabilistic modelling, simulation, and AI into applied software solutions.

Teaching Experience

Queen's University, School of Computing

Kingston, Ontario

Teaching Assistant

2023–Present

Courses: CISC 223 Software Specifications; CISC 452 Neural and Genetic Computing; CISC 352 Artificial

Intelligence; CISC 203 Discrete Structures II; CISC 151 Elements of Computing with Data Analytics.

- Supported instruction, assessment, student problem solving, and office-hour consultation across undergraduate computing courses.
- In CISC 352, developed content to teach causal reasoning and causal inference concepts.

Trent University

Guest Lecturer

Peterborough, Ontario

2024

- Delivered an undergraduate guest lecture on time-series forecasting.

Lakehead University

Graduate Assistant

Thunder Bay, Ontario

2020–2022

Courses: COMP 5421 Deep Learning; COMP 5012 Big Data; COMP 5013 Pattern Recognition; COMP 4111 Clinical Decision Support; COMP 2412 Data Structures.

- Marked assignments, labs, and examinations; held office hours; and prepared and delivered lectures and tutorials.

Presentations and Scholarly Dissemination

- 3rd International Conference on Foundation and Large Language Models (FLLM), Vienna, 2025. *Agent-guided Causal Discovery with a Small Language Model*.
- IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2023. Peer-reviewed conference publications in infectious-disease forecasting and hospital mental-health admissions forecasting.

Scholarships and Awards

- R. Samuel McLaughlin Fellowship, Queen's University, 2025–2026.
- Ontario Graduate Scholarship, 2024–2025.
- R. Samuel McLaughlin Fellowship, Queen's University, 2023–2024.
- Ontario Graduate Scholarship, 2021–2022.

Academic Service and Community Engagement

- **Technical Program Committee Member**, 4th International Conference on Foundation and Large Language Models (FLLM 2026), 2026. Reviewed submitted research papers and contributed to the technical program and peer-review process.
- **Reviewer**, *Information Processing & Management*, 2026.
- **Reviewer**, *INFOR: Information Systems and Operational Research*, 2026.
- **Ph.D.-Community Initiative**, Queen's University, 2023–2024. Collaborated with an interdisciplinary team and KFL&A Public Health to develop guidance for the responsible use of generative AI, including recommendations for practical and ethical adoption.

Selected Open Research Software

CaReBayes

github.com/bdmossop/CaReBayes

- Open-source implementation of the CaReBayes framework for regime-aware causal Bayesian forecasting of non-stationary time series.
- Includes workflows for temporal causal discovery, regime-specific Bayesian autoregressive modelling, regime classification, one-step-ahead forecasting, posterior predictive uncertainty quantification, and interventional forecasting.
- Provides reproducible experiments on simulated systems and real-world COVID-19 data corresponding to the PLOS ONE study, including benchmarking, evaluation metrics, and forecasting visualizations.

- Designed to support reproducibility and further research on causal forecasting under structural and regime change.

Technical Expertise

Causal inference	Structural causal models, DAGs, temporal causal discovery, constraint-based causal discovery, d-separation, interventions, causal feature selection.
Bayesian and probabilistic ML	Bayesian regression, posterior predictive inference, uncertainty quantification, probabilistic time-series models, PyMC.
Machine learning	Deep learning, time-series forecasting, convolutional neural networks, small language models, model evaluation.
Programming and tools	Python, C, C++, Java, JavaScript, SQL, MATLAB, R, PyTorch, JAX, NumPy, pandas, scikit-learn, Matplotlib, GitHub, LaTeX.
Software	AWS, Microsoft Azure, React, Django, Bootstrap.

Additional Industry Experience

TBayTel <i>Application Developer</i>	Thunder Bay, Ontario <i>Apr. 2019–Sept. 2019</i>
• Developed e-commerce web applications, graphical user interfaces, SQL-backed functionality, and supporting web services for telecommunications platforms.	
Terracon Geotechnique <i>Civil Engineer-in-Training</i>	Fort McMurray, Alberta <i>2017–2018</i>
• Conducted geotechnical field work including foundation and pile inspection, earthwork monitoring, materials testing, and field instrumentation.	
Opus Stewart Weir <i>Project Engineer</i>	Grande Prairie, Alberta <i>2015–2016</i>
• Supported and led transportation infrastructure projects involving proposals, design review, budgeting, field testing, survey coordination, construction oversight, and client communication.	
Thurber Engineering Ltd. <i>Field Engineer</i>	Edmonton, Alberta <i>2014–2015</i>
• Performed construction monitoring, materials testing, shallow and deep foundation inspection, and geotechnical analysis for projects across Alberta.	

References

Available upon request.